

Comparative Analysis of COVID-19 Data With Trends and Impacts in the Pre- and Post-Vaccination Periods of 2021

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Introduction

Bao Khoi Bui & Aufstin Filiko

With the introduction of the COVID-19 outbreak towards the end of 2019, there have been numerous infections, hospitalizations, mortalities, etc.. That, however, has been on a decline since the release of numerous vaccines for the disease around 2021. Being able to understand certain patterns and trends of COVID-19 before and after the emergence of vaccines through certain features can help decision makers identify factors that risk infection and mortality and help allocate resources to hospitals in need.

The goal of this project using the COVID-19 dataset provided is to answer the following questions:

1. Which features correlate with the spread and mortality of COVID-19?
2. How do vaccination rates influence trends in COVID-19 cases and mortality?
3. What factors are most predictive of hospitalization and ICU resource needs?
4. What patterns emerge from outliers in infection, hospitalization, and mortality rates?
5. How do socioeconomic and healthcare disparities affect pandemic outcomes?

Methodology

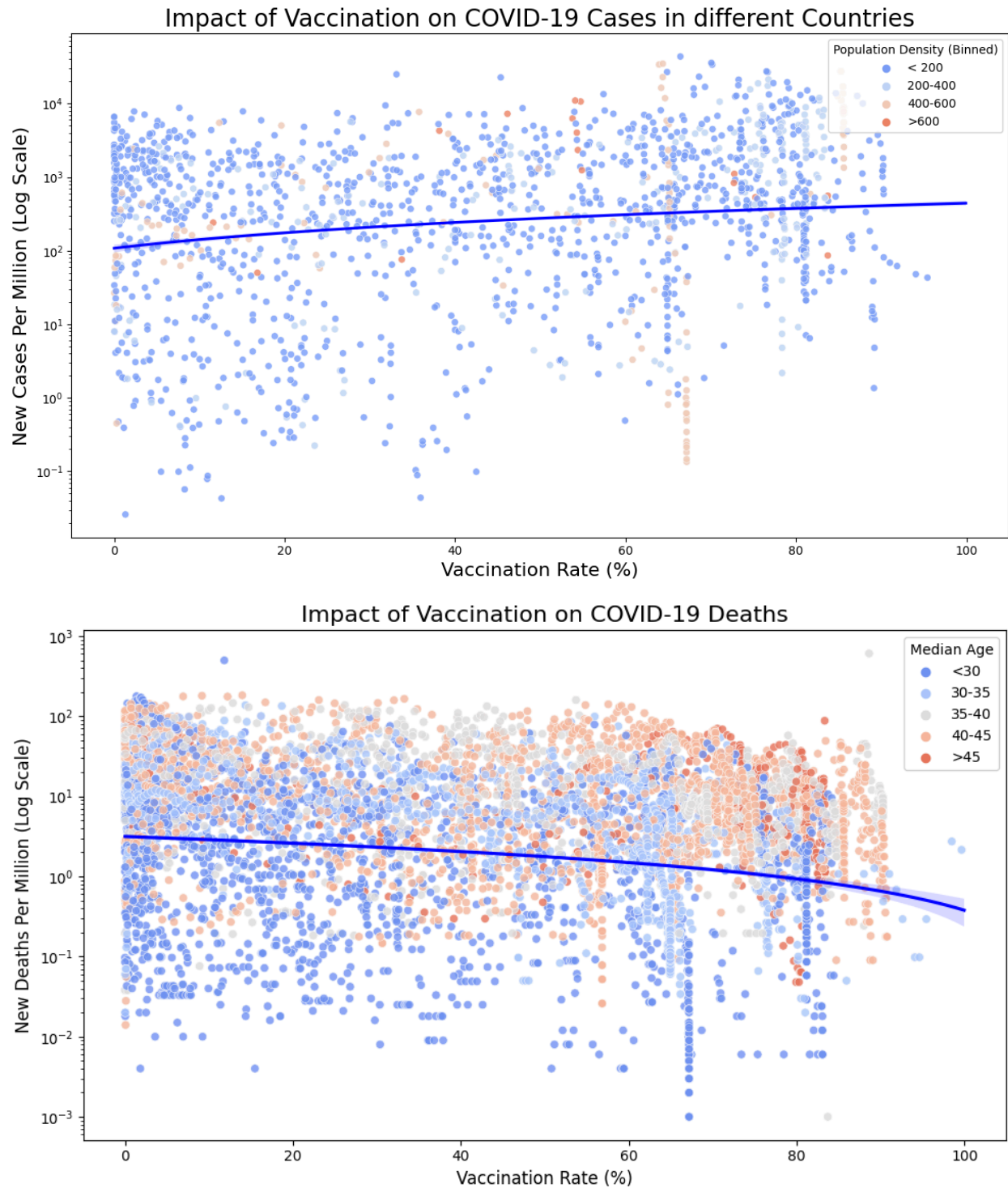
In this report, we first started with Exploratory Data Analysis (EDA) to visualize trends and correlations among key variables such as vaccination rates, hospitalization, mortality and GDP guide the direction of our future analysis. Data preprocessing involved categorizing regions into risk levels based on hospitalization and ICU rates. Classification models, including decision trees and random forests, were implemented to predict hospitalization needs and evaluate changes in healthcare resource requirements before and after vaccine availability. Finally, outlier detection methods, such as Z-Score and IQR, were used to identify and contextualize unusual spikes in COVID-19 metrics.

Exploratory Data Analysis (EDA) and Visualization

Shiv Bardolia & Aufstin Filiko

In the provided dataset with global COVID-19 data, many features referenced different points in time of the same country. Since this meant country data would have repeating data points of the same value and would heavily skew the resulting graphs, it was important to first group the data of a given feature in bins if it was going to be graphed. Some integer value ranges were very large even after initial normalization, such as in the number of cases per million, so it would further have to be altered in order to be better visualized. Regression lines were added to assist in visualizing possible relationships between many variables at once.

1. Impact of Vaccination on Covid-19 Cases and Deaths

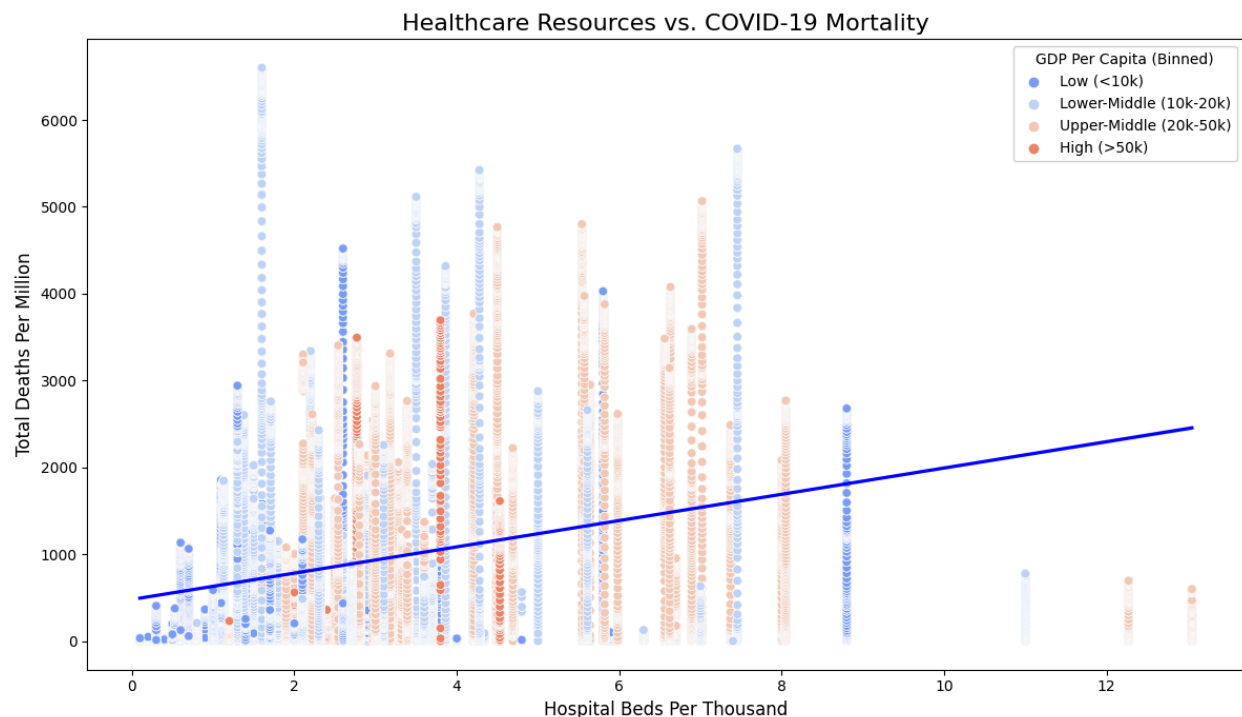


- The first plot illustrates the relationship between the percent of people vaccinated within a country at various points throughout the pandemic (**Vaccination Rate**) and the number of **New COVID-19 cases per million**, with points color-coded by bins with the median age of the country.

- Based on the the regression line, there appears to be a very weak positive correlation between vaccination rate and the number of new cases, which is a counter intuitive result.
- There does not appear to be a significant effect of population density on the rate of the number of cases based on the chart.
- The second plot illustrates the relationship between the percent of people vaccinated within a country at various points throughout the pandemic (**Vaccination Rate**) and the number of **New COVID-19 deaths per million**, with points color-coded by bins with the median age of the country.
- There appears to be a weak, negative relationship between vaccination rate and the number of new deaths, which would be expected. However, there also appears to be slightly more likely more deaths per million for countries with an older population compared to those with younger populations.

Due to the unexpected positive correlation between vaccination and the number of new cases, further analysis of the periods before and after the vaccine was available could help in verifying if it truly caused an increase in infection rate or if there are other variables involved.

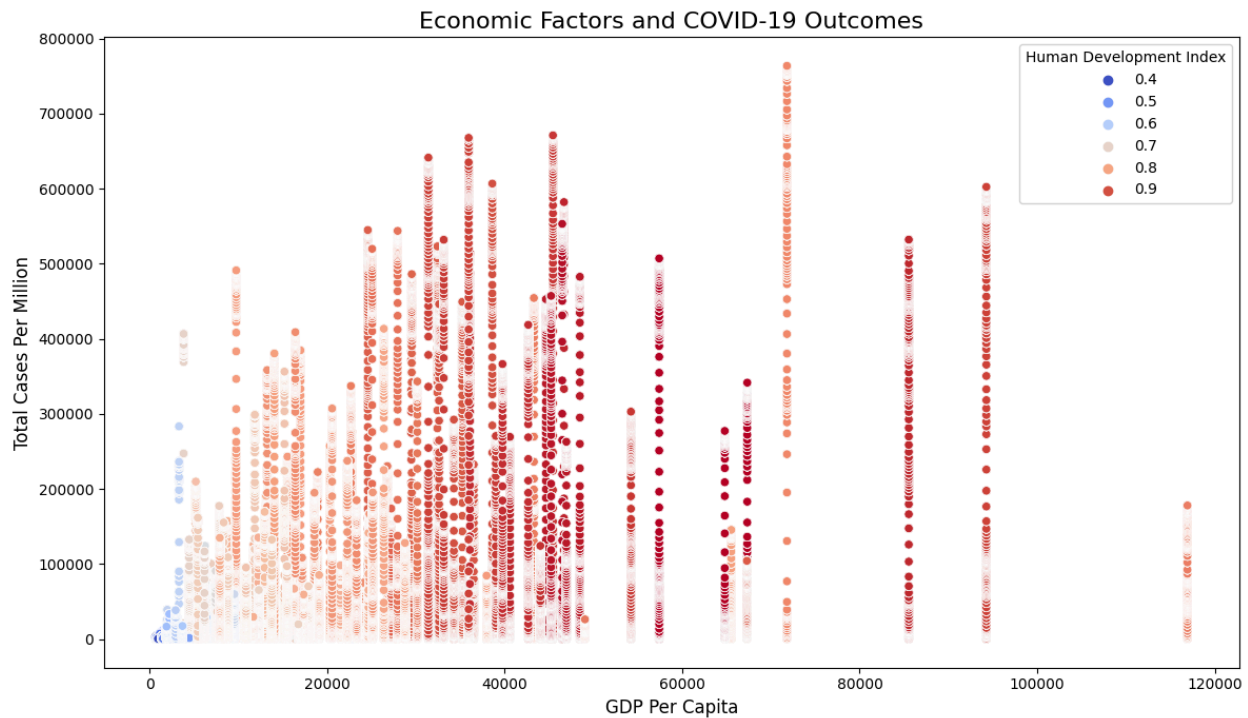
2. Healthcare Resources vs. COVID-19 Mortality



- The plot illustrates the relationship between the average number of hospital **beds per thousand** (a measure of healthcare capacity within a country) and **total deaths per million** (a measure of COVID-19 mortality), with points color-coded by GDP per capita.
- Due to the majority of countries having between 2-8 hospital beds per thousand people, as well as some countries with a much higher amount, the regression line appears to fit in a way that implies a moderate correlation between the number of hospital beds per thousand people and the amount of total deaths per million

- There is also a relationship with richer countries having an unforeseen higher mortality rate than countries with lower GDP, as those with lower GDP predictably have lower amounts of hospital beds.
- A wide variability in cases across all GDP per capita suggests that non-economic factors like pandemic response management and health policies play a role.

3. Economic Factors and Covid-19 Outcomes



- The plot examines the relationship between **GDP per capita** and **total COVID-19 cases per million**, with points color-coded by the Human Development Index (HDI).
- Both higher GDP per capita and HDI per capita appear to correlate with higher reported COVID-19 cases per million, likely due to increased testing and reporting capacity in wealthier countries. However, this is likely simply due to the correlation between HDI and GDP per Capita, as the distribution is similar to the graph of Healthcare Resources vs. COVID-19 Mortality.
- Outliers with extremely high cases per million exist in mid-to-high GDP and HDI regions, likely linked to specific outbreaks or delayed public health interventions.

There is more vertical stacking of values across the x axis due to the provided data being based around the tracking of the number of covid cases over time. Thus, the information from one country would appear as a column on the previous two graphs. While GDP and HDI show apparent correlations with total cases and mortality rates, the wide variability in outcomes across similar GDP levels indicates that other factors, such as pandemic response management, healthcare policies, and social behavior, may play more significant roles. Therefore, with some data pre-processing, we are more likely to obtain a better understanding of the data.

Data Pre-processing

Shiv Bardolia

The goal of the Data Pre-processing was to categorize COVID-19 risk levels into three categories **Low**, **Medium**, and **High** based on hospitalization rates, and ICU admissions. This classification aimed to provide a more interpretable assessment of pandemic severity across different regions.

```
Risk Level Summary:
risk_level
Low      10428
High     10423
Medium   7507
Name: count, dtype: int64
```

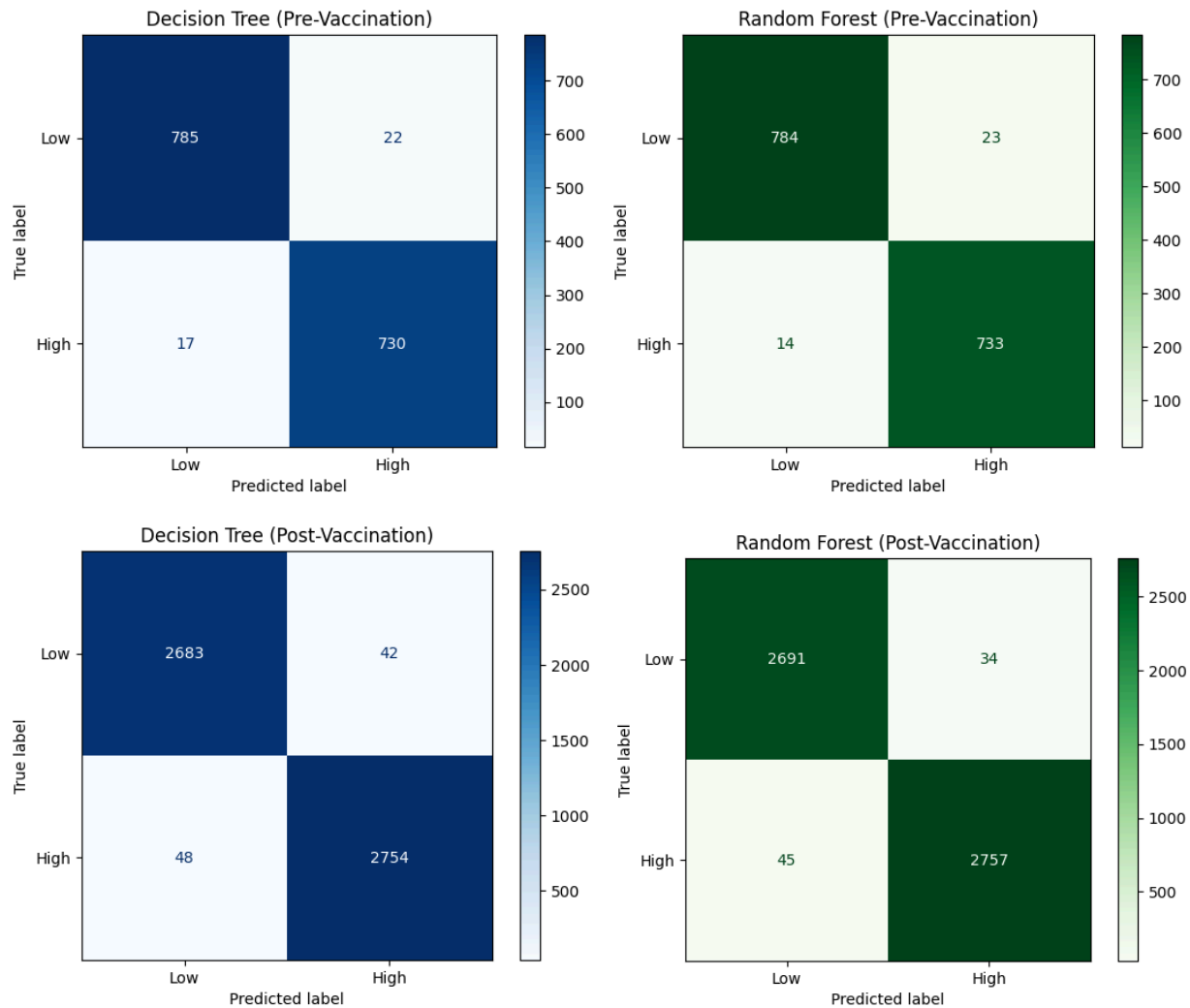
- **Low Risk:** 10,428 regions or observations were categorized as having low risk. These regions showed minimal hospitalization rates and ICU admissions, indicating effective containment or lower COVID-19 severity.
- Regions in this category can focus on maintaining existing healthcare policies while monitoring for potential outbreaks.
- **Medium Risk:** 7,507 regions fell into the medium-risk category. These areas displayed moderate hospitalization rates and ICU admissions, suggesting the need for ongoing monitoring and targeted healthcare interventions.
- Areas require targeted interventions, such as enhancing healthcare capacity or implementing stricter preventive measures.
- **High Risk:** 10,423 regions were identified as high risk. These regions experienced elevated hospitalization and ICU rates, highlighting critical pressure points in the healthcare system.
- Immediate resource allocation and policy focus are needed to address overwhelmed healthcare systems and reduce risk levels.

Classification

Bao Khoi Bui

Due to the results in our exploratory analysis, we want to determine which features are most significant in driving hospitalization and which of the following classification models are the best in predicting high hospitalization. By doing this, we are able to see which periods (pre-vaccination) and (post-vaccination) will require the most resources such as ICU beds, staff, vaccines, etc.

The following confusion matrices below shows the amount of hospitalization in two categories (low and high) in both vaccination periods (pre and post).



With these matrices, we find that:

- In terms of predictive ability, both classification models are very good and around the same level in predicting high hospitalization.
- The post-vaccination period matrix shows that there was nearly 4 times more true hospitalization compared to the pre-vaccination period.

In general, hospitals after the release of the covid vaccine will need more resources such as ICU beds, staff members, and vaccines.

The most significant features in predicting hospitalization rate were:

- icu_patients
- hosp_patients
- new_cases
- total_deaths

To get an even better idea of how effective the two classification models are, we measured the following metrics of accuracy, precision, recall, macro- and micro-f1 scores and is shown in the tables below in different vaccination periods:

Pre-Vaccination		
Metric	Decision Tree	Random Forest
Accuracy	0.974903	0.976190
Precision	0.974774	0.976016
Recall	0.974990	0.976379
Macro-F1	0.974872	0.976165
Micro-F1	0.974903	0.976190

Post-Vaccination		
Metric	Decision Tree	Random Forest
Accuracy	0.983716	0.985707
Precision	0.983701	0.985685
Recall	0.983728	0.985731
Macro-F1	0.983714	0.985704
Micro-F1	0.983716	0.985707

Result: While the random forest model is slightly, consistently better than the decision tree model, both models in general are highly effective in predicting hospitalization rate. The high predictive accuracy suggests that hospitalization trends are likely to follow predictable patterns based on key features in the data set.

Outlier Detection

Bao Khoi Bui

The objective of this method using Z-Score and IQR is to be able to detect periods with unusual high rates of infection, hospitalization, and mortality rates. We want to be able to also distinguish certain dates within the pre-vaccination and post-vaccination periods with certain events in the real world that may have an effect or influence in those unusual spikes. Then, we want to compare and analyze Z-Score and IQR detection methods between the two periods to see which one provides the best results. So, for comparison:

Outliers detected for infections:

```
Number of Z-Score Outliers - Pre-Vaccination: 46
Number of Z-Score Outliers - Post-Vaccination: 235
Number of IQR Outliers - Pre-Vaccination: 1119
Number of IQR Outliers - Post-Vaccination: 4025
```

Outliers detected for hospitalization:

```
Number of Z-Score Outliers - Pre-Vaccination: 128
Number of Z-Score Outliers - Post-Vaccination: 408
Number of IQR Outliers - Pre-Vaccination: 1044
Number of IQR Outliers - Post-Vaccination: 3527
```

Outliers detected for mortality:

```
Number of Z-Score Outliers - Pre-Vaccination: 78
Number of Z-Score Outliers - Post-Vaccination: 214
Number of IQR Outliers - Pre-Vaccination: 995
Number of IQR Outliers - Post-Vaccination: 3840
```

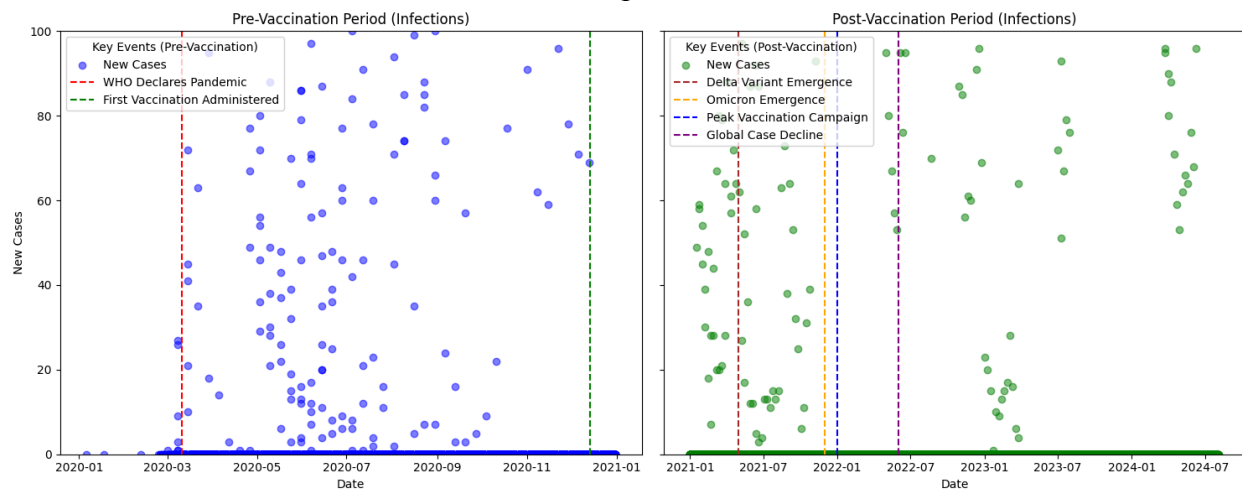
As we can see, there are a few things to note when looking at the number of outliers detected by the Z-Score and IQR detection methods:

1. Z-Score captures less outliers in both pre- and post-vaccination periods compared to IQR.
2. Post-vaccination outliers are captured more than pre-vaccination in both detection methods.

This means there are more infections, hospitalizations, and mortality rates after vaccination began than before vaccination became a thing. Explanations for this could be:

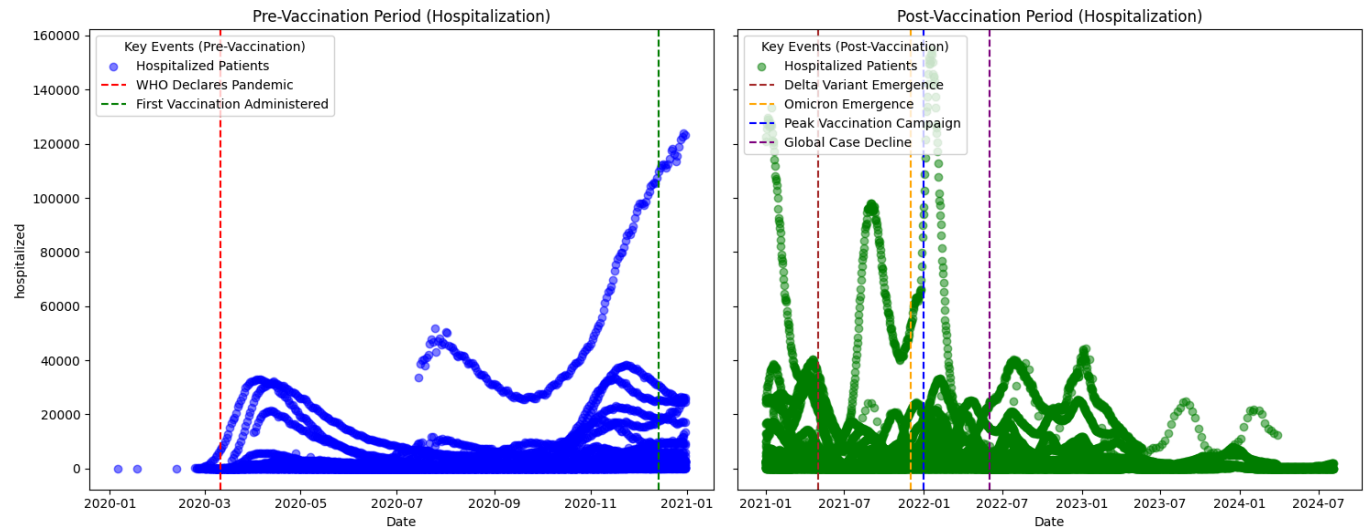
1. There might have been some populations out there around the world on certain dates/times that have yet to be vaccinated.
2. Vaccinations could have not turned out well for certain regions in the world.
3. Reporting of these metrics might have been delayed and improved which could cause an unusual spike.
4. Certain events such as the introduction to the Omicron and Delta variants could lead to an instant rise in the metrics.

For the following scatterplots, we can see when certain events happen at different parts of time within the Pre-vaccination and Post-vaccination periods.



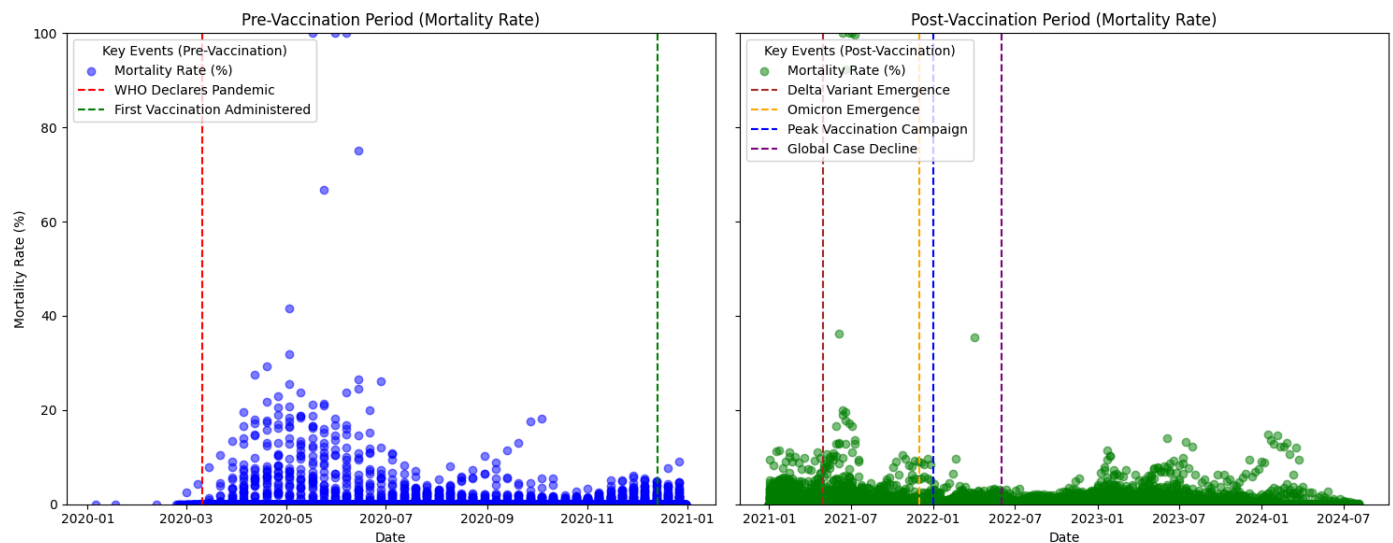
Based on the infection scatterplot we see that:

- Regardless of any event, there are still several outliers in all parts of the pre- and post-vaccination periods.
- There does not appear to be a large difference in the number of new cases before or after the vaccine was introduced.



Based on the hospitalization scatterplot we see that:

- This supports the findings on how there are more outliers in the post-vaccination period due to certain events such as the emergence of the Omicron and Delta variants.
- There is a steep drop in hospitalized patients when the vaccine was first administered, which while it didn't reduce the number of outliers, it did show the vaccine's effectiveness in slowly reducing patient hospitalization until there appears to be a few countries that had large spikes in hospitalization even after the vaccine was created.



Based on the mortality rate scatterplot, we see that:

- There are a few outliers that could be the result from days in which many deaths were counted at once or countries with varying reporting policies that coincided with spikes in infections and mortality.
- There was a decrease in mortality after an initial spike during the first outbreak of COVID-19 as well as smaller spikes in mortality throughout both periods.
- There is a drop in mortality rate in the post-vaccination period compared to the pre-vaccination period, which shows that the vaccine could be effective against the virus.

Conclusion

Aufstin Filiko

Working with the provided COVID-19 pandemic data, our team was able to implement data analysis techniques and methodologies to visualize and uncover patterns and relationships in the pre- and post-vaccination periods of 2021. The exploratory analysis initially provided an unexpected, albeit weak, positive correlation between vaccination rate and the number of new cases. This led us to further analyze the pre- and post- vaccination periods to better understand the vaccine's effect on the number of new cases and deaths.

We also realized many confounding factors play a role in the spread of severity of COVID-19, so we attempted to find the best model for predicting hospitalizations based on the provided data. Afterwards, we performed outlier detection through the use of normalization of z-scores and applying IQR to the data. Although the decision tree and random forest models did not produce drastically different results, their consistent performance validated the dataset's reliability and confirmed there were relationships to be found in the data that could be used to create reliable models.

We compared the graphs based on vaccination rate once again and found that the global access to vaccines had a visible effect on reducing the number of new cases, decreasing mortality rate, and lowering the number of hospitalizations over time. Although there were still discrepancies in the data as countries decreased their respective recordings of COVID-19 data, there is still evidence to suggest that the vaccines were effective in combating the deadly effects of the pandemic.

Our team could have processed the data further in order to prevent the stacking of data points on top of each other when trying to display country-wide data. It could be possible to use the same time stamps provided by the source of the data as a way to reduce the noise when comparing things between countries. Nevertheless, this analysis can serve as a foundation for further exploration of individual variables across the globe that have a greater impact on pandemic outcomes.